Hieu Nguyen

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Professional Summary

Software engineer and technical lead with 7+ years of experience modernizing enterprise systems and leading full-stack development efforts. Specialized in cloud migrations, automation, backend architecture, and AI-enhanced solutions while working with Verizon (via TCS). Delivered major cost savings, security improvements, and efficiency gains while helping secure new project funding through innovative approaches. I build scalable, end-to-end systems that deliver real business value.

Technical Skills

- **Languages:** Java (8/11/17), Python, TypeScript, JavaScript, SQL, Groovy, XSLT, Go
- **Cloud & DevOps:** AWS (Lambda, EC2, S3, CloudWatch, CloudFormation), Terraform, Kubernetes, Docker, GitLab CI/CD, Jenkins
- **Frameworks & Architecture:** Spring Boot, Spring Cloud, Spring Batch, Hibernate, Microservices, Mediator Pattern, RESTful APIs, Event-Driven Systems
- **Frontend:** Angular (2+), React, HTML5, CSS3
- **Databases & Messaging:** PostgreSQL, Cassandra (NoSQL), MySQL, Apache Kafka, Apache Pulsar
- **Testing & Security:** JUnit, Mockito, Cucumber, Selenium, JMeter, SonarQube, Fortify, Black Duck
- **AI Tools:** LangGraph, LangChain, RAG, GitHub Copilot
- **Methodologies:** Agile (Scrum, Kanban), SAFe

Professional Experience

Technical Lead / Full Stack Developer
Tata Consultancy Services (Client: Verizon) — Irving, TX

May 2019 – Present

- Refactored monolithic qualification platform with a custom **Mediator orchestration pattern** (**Java, Groovy**) to break down complex processes into discrete, maintainable steps.
- Drove **50%** reduction in **EC2** infrastructure costs by optimizing non-production environments: scaled **SIT** from multiple pods to **one pod** per service, archived logs from **OpenSearch** to **S3**, and shut down environments on weekends.
- Directed modernization of **30+** enterprise services from **Java 8** to **Java 17** and **Spring Boot 2 to 3**, resolving **90+** critical security vulnerabilities.

- Performed targeted performance tuning on key **APIs** and batch processes. For example, reducing one critical batch import from **2 hours to 10 minutes** and improving a high-traffic API response time from **2 seconds to 250ms**.
- Co-designed a multi-agent **AI system** (**LangGraph + RAG**) with a supervisor agent orchestrating specialized agents for intent extraction, tool calling, business logic analysis, and action execution.This natural language service qualification assistant replaced manual workflows and added demand-based inventory insights for network profitability decisions, directly contributing to a **$500K** project win.
- Automated the CPQ (Cost Price Quote) workflow, replacing a manual process involving emails, Slack, and Excel with an application-driven flow – saving an estimated $80,000 per year.
- Created an automated testing suite with **Cucumber and Selenium**, reducing manual regression testing from **2 days to 4 hours** while increasing coverage by **80%**.
- Partnered with architecture teams to implement standardized **CI/CD pipelines** and **TDD/BDD** practices across multiple teams, reducing deployment failures by **25%**.
- Owned end-to-end feature delivery: gathered requirements from Product Owners, analyzed current systems, designed solutions (including **POCs**), and secured stakeholder alignment.
- Led team execution through daily standups, blocker resolution, and code reviews. Created feature analysis documentation with acceptance criteria and processing algorithms to reduce bugs and keep the team aligned.

PERSONAL PROJECTS

Quantitative AI Portfolio Platform

- Architected a 12-stage investment pipeline using LangGraph, combining LLM market sentiment analysis with deterministic models (Kelly Criterion) for trade sizing.
- Built a real-time data ingestion pipeline and WebSocket communication between FastAPI backend and React frontend for sub-second portfolio updates.

AWARDS & RECOGNITION

- **Verizon Hackathon – 4th Place (out of 30+ teams):** Developed a code generation tool that converts declarative JSON UI designs into functional JavaScript applications.
- **TCS Corporate Awards:** Technical Excellence (Feb 2025), Star of the Quarter (May 2024), Best Team Award (Oct 2023).

Education

Bachelor of Science, Biochemistry, Texas Christian University — Fort Worth, TX

Associate of Science, Computer Science, Tarrant County College — Fort Worth, TX